

ECE496Y - Budget											
Project ID #: 2025162											
Project Title: Vision-Guided Automated Foosball Opponent											
Students: Jonathan Ryu, Luke Watson, Daniel Yu, William Zhang											
Supervisor: Raviraj Adve											
Date	Purchase #	Quantity/Component	Price (\$CAD)	Purpose	Receipt						
06-14-2025	1	2x Ball	\$14.68	Testing software ball detection	20250713_10391						
07-08-2025	2	1x Miniture Foosball Tables	\$72.30	Modelling on small scale design	Small foosball ta						
07-09-2025	3	1x Arducam 100fps Global Shutter Color USB Camera Board	\$83.29	Camera for foosball table vision input	Amazon_order_2						
08-20-2025	4	1x 3D Print	\$2.48	Camera housing structure testing	Screenshot 2025						
08-28-2025	5	1x 3D Print	\$23.97	Camera housing structure testing	Screenshot 2025						
09-12-2025	6	1x iCAN USB-A 2.0 Male to RS232 DB9 Adapter	\$11.29	Cable adapter to program settings of servos from PC	20250915_18250						
09-15-2025	7	1x HPVB45 400mm Timing Belt Linear Slide Guide	\$152.58	Linear movement system for foosball rod	Sept. 15, 2025 –						
09-15-2025	8	2x NEMA 23 Integrated Easy Servo Motor 180W 3000rpm 0.6Nm	\$348.52	Servos for rotational and linear movement	Sept. 15, 2025 –						
09-16-2025	9	1x DHL Import Duty and Tax (StepperOnline)	\$57.65	Import tax payment for #8	Sept. 16, 2025 –						
09-18-2025	10	1x Raspberry Pi 5 Single Board Computer 2.4GHz 4 Core 16GB RAM 1x Raspberry Pi 5 Active Cooler 1x 27W USB-C PSU	\$226.08	Computer that will process camera input and perform all computations	DigiKey - Print.p						
09-29-2025	11	1x Foosball Table	\$80.00	The base table	Sept. 29, 2025 –						
10-03-2025	12	1x Lexar 128GB microSDXC Card	\$19.20	Storage for #10	CustomerInvoice						
10-05-2025	13	2x LRS-350-36 Power Supply 1x Pi 5 Case	\$134.50	Power supplies for #8	Oct. 5, 2025 – P						
10-04-2025	14	1x Spider Coupler	\$18.73	Test coupler for foosball rod	Coupler						
10-04-2025	15	1x Power Cord 1x Wire Connectors	\$70.08	General purpose wire terminals and power for (13)	Cord+Connector						
10-04-2025	16	1x Connector Plug	\$39.20	Inlets for power cords to power supplies	Connector plug.p						
10-09-2025	17	1x Jett Challenge Foosball Rod 3 Hole 5/8in	\$19.99	Foosball rod to be actuated on the table	invoice-75206.p						
10-11-2025	18	1x Electrical Wires	\$18.06	General purpose wires for power and data transmission	Electrical wires.p						
10-29-2025	19	1x Gloves 1x Pack of Nails 4x 2"x2"x8"	\$28.19	Tools and wood for building camera housing structure	20251102_11383						
10-29-2025	20	18x L brackets	\$12.76	Connecting the camera housing structure	L-brackets.pdf						
10-31-2025	21	24x Screws	\$1.68	Tools for building camera housing structure	스크린샷 2025-1						
11-01-2025	22	2x 8mm x 5/8" Flexible Jaw Spider Shaft Coupling	\$57.47	Couplers sized for foosball rod and servo shaft	srcdoc.pdf						
11-01-2025	23	1x Teknic ClearCore Controller	\$179.60	Device that will control all servos, programmed with #10	Gmail - Teknic, Ir						
11-02-2025	24	1x LRS-150-24 Power Supply	\$37.00	Power supply for #23	Nov. 2, 2025 – L						
11-03-2025	25	1x Power Cord 1x Power Socket 1x 66 ft 14AWG Wire	\$74.46	Wires for power delivery and accesories for #24	order13941.pdf						
11-03-2025	26	18x L-brackets 1x Power cord	\$34.61	Connecting the camera housing structure and power for #13	cord						
11-06-2025	27	1x FedEx Shipping Duties	\$32.07	Duties for #23	Gmail - FedEx P						
11-09-2025	28	4x 0395105003 connector 5x 0039039082 connector 25x 0039000047 crimp 35x 1181050 crimp	\$36.04	Connectors for #23	Gmail - FedEx P						

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